1 A parabola, P with equation $(y-a)^2 = ax$, where a is a constant, undergoes in succession, the following transformations:

A: A translation of 2 units in the positive x-direction

B: A scaling parallel to the y-axis by a factor of $\frac{1}{3}$

The resulting curve, Q passes through the point with coordinates $\left(2, \frac{4}{3}\right)$.

(i) Show that $a = 4$. [3]

(ii) Find the range of values of k for which the line $y = kx$ does not meet P. [3]

2 The region bounded by the curve $y = \frac{1}{\sqrt{x}-2}$, the x-axis and the lines $x = 9$ and $x = 16$ is rotated through 2π radians about the x-axis. Use the substitution $t = \sqrt{x}$ to find the exact volume of the solid obtained. [6]

3 **(i)** Express $\frac{r+1}{(r+2)!}$ in the form $\frac{A}{(r+1)!} + \frac{B}{(r+2)!}$, where A and B are integers to be found. [2]

(ii) Find $\sum_{r=1}^{n} \frac{r+1}{3(r+2)!}$. [3]

(iii) Hence, evaluate $\sum_{r=0}^{\infty} \frac{r+1}{3(r+2)!}$. [2]

4 Kumar wishes to purchase a gift priced at $280 for his mother.

Starting from January 2017,

- Kumar saves $100 in his piggy bank on the 1st day of each month;
- Kumar donates 30% of his money in his piggy bank to charity on the 15th day of each month and
- Kumar's father puts an additional $20 in Kumar's piggy bank on the 25th day of each month.

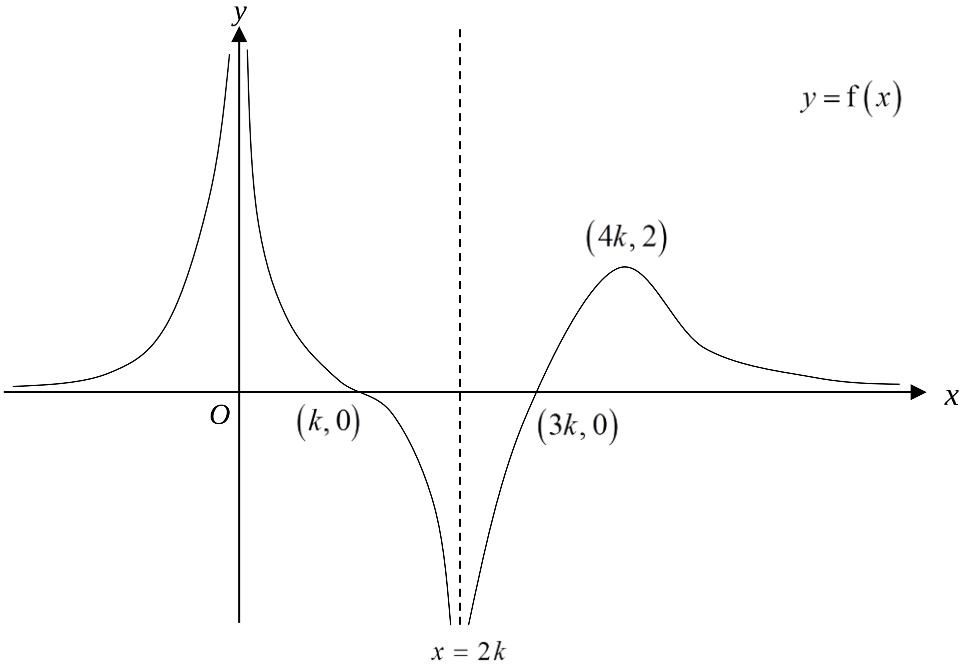
(i) Find the amount of money in Kumar's piggy bank at the end of March 2017. [2]

(ii) Show that the amount of money in Kumar's piggy bank at the end of n months is

$$300\left(1-0.7^n\right).$$ [3]

(iii) At the end of which month will Kumar first be able to purchase the gift for his mother? [2]

5 The diagram below shows the sketch of the graph of $y = \mathrm{f}(x)$ for $k > 0$. The curve passes through the points with coordinates $(k, 0)$ and $(3k, 0)$, and has a maximum point with coordinates $(4k, 2)$. The asymptotes are $x = 0$, $x = 2k$ and $y = 0$.



Sketch on separate diagrams, the graphs of

(i) $y=\mathrm{f}(-x-k)$, [2]

(ii) $y=\mathrm{f}'(x)$, [2]

(iii) $y=\dfrac{1}{\mathrm{f}(x)}$, [3]

showing clearly, in terms of k, the equations of any asymptote(s), the coordinates of any turning point(s) and any points where the curve crosses the x- and y-axes.

6 A straight line passes through the point with coordinates $(4,\ 3)$, cuts the positive x-axis at point P and the positive y-axis at point Q. It is given that $\angle PQO=\theta$, where $0<\theta<\dfrac{\pi}{2}$ and O is the origin.

(i) Show that the equation of line PQ is given by $y=(4-x)\cot\theta+3$. [2]

(ii) By finding an expression for $OP+OQ$, show that as θ varies, the stationary value of $OP+OQ$ is $a+b\sqrt{3}$, where a and b are constants to be determined. [5]

7 A curve C has parametric equations

$$x=\frac{4}{t+1} \quad \text{and} \quad y=t^2-3, \quad t\neq -1.$$

(i) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t. [2]

(ii) Find the equation of the normal to C at P where $x = -2$. [3]

(iii) Find the other values of t where the normal at P meets the curve C again. [3]

8 The curve C has equation

$$y = \frac{2x^2 - 3x + 5}{x - 5}.$$

(i) Express y in the form $px + q + \frac{r}{x-5}$ where p, q and r are constants to be found. [3]

(ii) Sketch C, stating the equations of any asymptotes, the coordinates of any stationary points and any points where the curve crosses the x- and y-axes. [4]

(iii) By sketching another suitable curve on the same diagram in part **(ii)**, state the number of roots of the equation

$$\left(2x^2 - 3x + 5\right)^2 = 5x\left(x-5\right)^2.$$ [3]

9 **(a)** Given that the first two terms in the series expansion of $\sqrt{4-x}$ are equal to the first two terms in the series expansion of $p + \ln\left(q - x\right)$, find the constants p and q. [5]

(b)(i) Given that $y = \tan^{-1}\left(ax+1\right)$ where a is a constant, show that $\frac{\mathrm{d}y}{\mathrm{d}x} = a\cos^2 y$. Use this result to find the Maclaurin series for y in terms of a, up to and including the term in x^3. [5]

(ii) Hence, or otherwise, find the series expansion of $\frac{1}{1+\left(4x+1\right)^2}$ up to and including the term in x^2.

[3]

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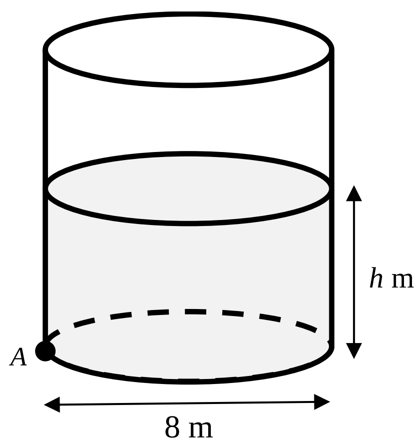


The figure above shows a cylindrical water tank with base diameter 8 metres. Water is flowing into the tank at a constant rate of $0.36\pi\, \text{m}^3$/min. At time t minutes, the depth of water in the tank is h metres. However, the tank has a small hole at point A located at the bottom of the tank. Water is leaking from point A at a rate of $0.8\pi h$ m^3/min.

(i) Show that the depth, h metres, of the water in the tank at time, t minutes satisfies the differential equation

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1}{400}(9-20h).$$ [3]

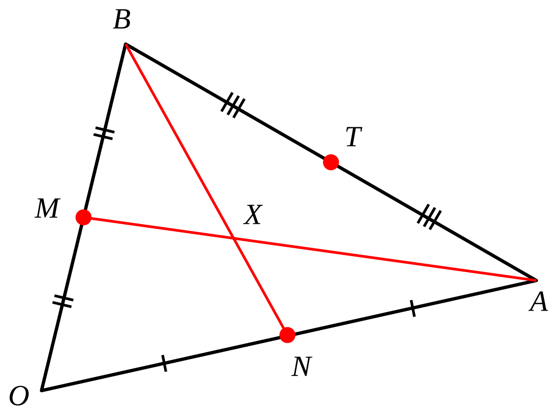
(ii) Given that $h = 0.4$ when $t = 0$, find the particular solution of the above differential equation in the form $h = \mathrm{f}(t)$. [6]

(iii) Explain whether the tank will be emptied. [1]

(iv) Sketch the part of the curve with the equation found in part **(ii)**, which is relevant in this context. [2]

11 A median of a triangle is a line segment joining a vertex to the midpoint of the opposite side.

For the triangle shown below, O, A and B are vertices, where O is the origin, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The midpoints of OB, OA and AB are M, N and T respectively.



It is given that X is the point of intersection between the medians of triangle OAB from vertices A and B.

(i) Show that $\overrightarrow{OX} = \frac{1}{3}(\mathbf{a}+\mathbf{b})$. [4]

(ii) Prove that X also lies on OT, the median of triangle OAB from vertex O. [2]

The **centroid** of triangle OAB is the common point of intersection X between all three medians of the triangle.

Ray tracing is a technique in computer graphics rendering used to realistically capture the lighting effect in a scene being modelled. Starting from a chosen viewpoint, different rays are being traced backwards towards different parts of an object in the scene and reflected off the object. For each ray, if it reflects off the object and intersects a light source, then the part of the object at which the ray is reflected off would be made to appear brighter.

In a particular scene depicting a dolphin jumping out of the ocean, a ray is being traced back from a chosen viewpoint at V to the **centroid** X of a particular triangular facet defined by the vertices comprising the origin O, $A(5,\ 4,\ 6)$ and $B(-2,\ 2,\ 3)$, and then reflected off the facet at X, as shown in Figure 1.

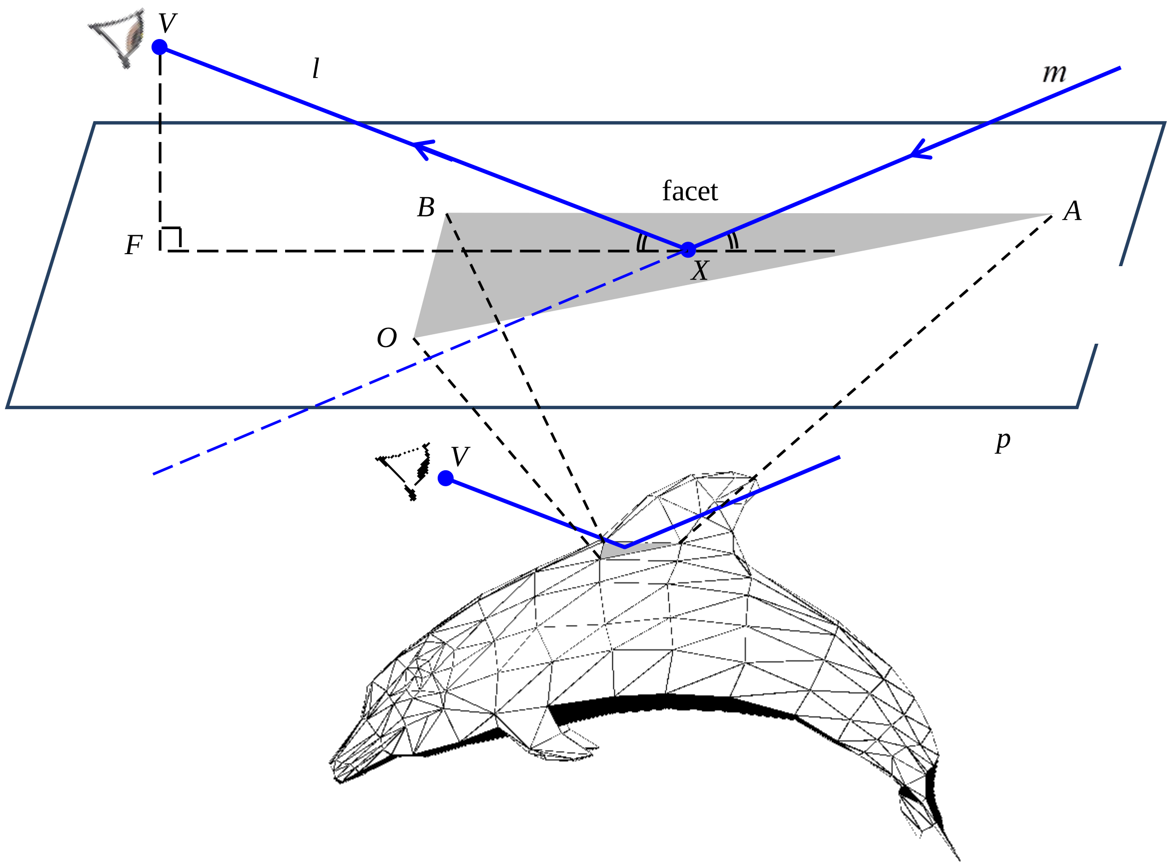


Figure 1

(iii) Show that the plane p which contains the triangular facet OAB can be represented by the cartesian equation $-3y+2z=0$. [2]

(iv) Given $V(1,-68,-37)$, determine the coordinates of the foot of perpendicular F from V to plane p. [4]

The reflected ray travels along a line m such that:

- both line VX (denoted by l) and line m lie in a plane that is perpendicular to plane p, and
- the angle between line l and plane p equals the angle between line m and plane p.

(v) By first finding two suitable points lying on line m, or otherwise, find a cartesian equation for line m. [5]